How Does Technology Affect our Relationship with God?

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1 Introduction

Within the field of Human-Computer Interaction (HCI), significant research has focused on designing systems, platforms, and content that capture and sustain user attention. Modern Smartphones employ persuasive design strategies, personalized algorithms, and continuous notifaction systems to maximize engagement. While these technologies have increased accessibility, communication, and productivity, they have also contributed to unprecedented levels of screen time and digital dependence.

As engagement with digital devices become more constant, researchers have begun examining the broader implications of sustained technological interaction. Much of the existing literature explores impact on mental health, productivity, and interpersonal relationships. However, comparatively less attention has been given to how these technologies influence spiritual life and religious practices, specifically in Christianity. From both psychological and theological perspectives, humans are often described as relational beings whose well being depends on intentional connection with others (e.g., Genesis 2:18), and for many, with God. Practices such as prayer, solitude, worship, and reflection require sustained attention and cognitive presence, qualities that are challenged by distracting environments.

This paper investigates the question: How does technology affect our relationship with God? This will be done by a literature review that examines research in HCI, Customer Smartphone Distraction(CSD), Technology in spiritual disciplines, Prayer, and Scripture reading. By synthesizing these findings, this review identifies emerging themes, points of agreement, and areas where further research can be done.

2 Background

To better understand the relationship between technology and spiritual life, it is important to examine key concepts that shape this discussion. The following section provides background on Human Computer Interaction, smartphone distraction, and Christian spiritual disciplines, establishing the conceptual foundation needed to evaluate exisiting research in these areas.

2.1 HCI

Human Computer Interaction is an interdisciplinary field concerned with the design, evaluation, and implementation of interactive computing systems for human use. A central area of inquiry within HCI examines how interface design influences user behavior and attention[2]. Design elements ranging from visual layout and button placement to notification systems and algorithmic content delivery shape how users engage with digital environments. While HCI research has traditionally emphasized usability, efficiency, and health applications, more recent work has begun exploring religion and spirituality as potential design spaces. However, these areas remain comparatively understudied within mainstream HCI scholarship.

When it comes to HCI, being human-centered is extremely important, in fact it's considered to be one of the pillars of HCI as a discipline [3]. But what does it mean to be human-centered? Human-centered design means that the choices being made in a design process are being made to tailor to the user's needs, goals, and preferences. This is done by understanding the user through research and then using that understanding to inform design decisions.

While it's important to be human-centered in design, in order to be human-centered, you have to understand how to do user research. User Research is the process of understanding the methods used to obtain actionable knowledge about users, including their activities, needs, wants, and contexts of use [3]. User Research stems from the realization that the designer and the user are not the same person and that the designer cannot assume to know what the user needs or wants without first understanding them. User research can be done in a variety of ways, including interviews, surveys, observations, and usability testing.

Another important concept in HCI is interaction. Interactions can describe many different things, like individuals, groups, and communities using computers. Interactions also take place in different timescales, like pressing a button, or onboarding employees to a new software system. To put it more generally, interaction is a dynamic phenomenon that unfolds over time as users and computers influence each other.

User Interfaces are the means through which users interact with computers. One source describes user interfaces as "the only part of a system that is used by real people doing real tasks in real settings. Therefore, what makes an interface good is ultimately determined by humans. As a result research on user interfaces is a cross-disciplinary effort with contributions from computer science, electrical engineering, design, and psychology" [3]. User interfaces can be designed in a variety of ways, but the goal is to create an interface that is efficient, and effective. Another goal for most interfaces is to make it easy to learn and use, but take a website like Bloomberg Launchpad for example, which is a financial news website. It is not designed to be easy to learn and use, but rather to be efficient and effective for its target audience, which is financial professionals. This is an example of how user interfaces can be designed with different goals in mind, and that the design of an interface can have a significant impact on how users interact with it.

Figure 1: Bloomberg Launchpad Interface

Design is also a central concept in HCI. Design can be described as the process of making choices about high level concepts or minute details to enhance the user experience. Design affects the way we perceive, experience, remember, decide, and form habits. Design can make or break someone's ability to use something, even small details can have significant impacts.

2.2 CSD

One important concept relevant to this discussion is Customer Smartphone Distraction, which is defined as an internet-connected smartphone competing for and compromising a user's attentional resources that were originally directed toward another task[7]. Mobile Devices are frequently engineered to maximize engagement through persistent connectivity, notifications, and personalized content feeds. As a result, they may fragment attention and interrupt sustained cognitive or reflective activities. Given that many spiritual practices require focused attention and intentional presence, this dynamic raises important interdisciplinary questions.

2.3 Spiritual Disciplines

In this paper, spiritual practices are discussed within a Christian framework. Spiritual discipline refers to structured activites intended to cultivate a relationship with God, including prayer, Scripture reading, worship, solitude, silence, and rest. Prayer has been defined as "every kind of inward communion or conversation with the divine power"[1]. A quote from Finney and Malony that defines prayer in a Christian context is, "In Christianity God is seen to be essentially all-loving and all-powerful, the one who has revealed himself in Jesus, The Christ. In the Christian tradition prayer is, this, intimate communication with a benevelont deity." This study of prayer suggests that such practices involve focused cognitive and emotional investment, further underscoring the importance of sustained attention in spiritual life.

Worship functions as a broader category encompassing various forms of devotional expression, including singing, praise, thanksgiving, and service. While these practices vary in format, they share an emphasis on intentional orientation toward the divine. As such, they provide a meaningful lens through which to examine how technological environments, particularly those optimized for interruption and engagement, may influence patterns of spiritual practice and relational presence.

2.4 Technology in Spiritual Disciplines

In addition to shaping patterns of attention, digital technologies also function as platforms through which spiritual practices may be facilitated. Mobile applications provide access to Scripture, guided prayer resources, devotional content, and livestreamed worship services, extending religious participation beyond traditional physical spaces[8]. For individuals with limited access to in-person communities, digital environments may enable forms of connection and engagement that would otherwise be unavailable. From an HCI perspective, this reflects the capacity of interactive systems not only to capture attention, but also to mediate meaningful experiences, including those related to faith and religious practice. As a result, technology's role in spiritual life is not solely disruptive, but complex and multifaceted.

Together, these perspectives suggest that technologies designed to capture and sustain attention may simultaneously enable and complicate practices that depend upon sustained relational presence. This tension forms the foundation for examining how technology may affect an individual's relationship with God.

3 Related Works

Having established key concepts surrounding engagement driven design, attention fragmentation, and Christian spiritual disciplines, the next section examines exisiting research that explores the intersection of these areas.

3.1 HCI

Research in HCI increasingly examines how interactive systems are intentionally designed to optimize user engagement and sustain attention. Studies within HCI highlight how interface elements such as notification systems, personalized content delivery, and feedback loops reinforce habitual interaction patterns and extend user engagement [2, 8]. These design strategies are frequently framed as enhancing usability, responsiveness, and user satisfaction by reducing friction and encouraging continued participation. Further work in persuasive and reflective design suggests that digital environments not only respond to user behavior but actively shape attentional habits through structured interaction flows and engagement incentives [4]. Together, this body of research positions attention optimization as a central principle in contemporary digital systems, establishing engagement as both a design objective and a behavioral outcome of interface architecture.

3.2 CSD and Cognitive Fragmentation

Empirical studies on smartphone distraction and cognitive fragmentation demonstrate that attentional disruption emerges from an interaction between device design features, psychological states, and environmental stimuli[7]. Research indicates that notifications, app design, and constant connectivity contribute to fragmented attention, reducing users' ability to sustain focus on tasks or prolonged cognitive activities. Similarly, studies examining technology's effect on Christian practice suggest that pervasive smartphone usage distract and take away from core religious values and reshape patterns of engagement with spiritual teachings. It has also been argued that habitual reliance on digital platforms can influence behavior and priorities, potentially altering the expression of Christian identity[6]. Together, these studies suggest that while smartphone design effectively captures and sustains user attention, this engagement may unintentionally disrupt activities requiring pronlonged focus, including spiritual practices. This tension between attention optimizing technologies and religious practices that depend on sustained focus underscores the need to examine how device mediated environments interact with religious and contemplative behaviors.

3.3 Spiritual Disciplines

Christian spiritual practices are traditionally understood as intentional activities that cultivate reflection, attentiveness, and relational engagment with God. Common disciplines include prayer, scripture reading, worship, silence, and solitude, all of which require sustained focus and intentional participation. Prayer, for example, has been described as a form of inward communion or conversation with the divine, emphasizing a relational interaction between the believer and God [1]. Within Christian theology, this interaction is often framed as intimate communication with a benevolent and personal deity, making attentiveness an essential component of the practice. Similarly, engagement with scripture involves interpretive reflection and critical understanding rather than passive consumption of information. Research on Bible engagement highlights the importance of thoughtful reading and hermeneutical interpretation in fostering spiritual understanding and personal transformation [5]. Together, these practices emphasize sustained attention and intentional reflection as central components of Christian spiritual life.

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3.4 Technology in Spiritual Disciplines

While concerns about distraction are significant, research also suggests that digital technologies can play a constructive role in supporting spiritual practices. Recent work in HCI has begun to explore religion and spirituality as potential spaces for thoughtful design, examining how digital tools can facilitate reflection, prayer, and engagement with religious communities [2]. Mobile applications, online worship services, and digital scripture platforms allow individuals to access religious resources more readily and maintain connection with faith communities across geographic boundaries. In some cases, these technologies provide new opportunities for spiritual engagement by integrating reminders for prayer, guided devotional content, or community discussion into everyday digital environments. Rather than

functioning solely as sources of distraction, these systems demonstrate that technology can also mediate and support spiritual practices when designed with reflective or contemplative goals in mind. As a result, the relationship between digital technology and spirituality is not purely oppositional but represents a complex interaction between engagement driven design and the intentional practices that characterize religious life.

3.5 Conclusion of Sources

Taken together, the literature reviewed above highlights a potential tension between the engagement driven nature of modern digital technologies and the attentional demands of Christian spiritual practices. Research in Human Computer Interaction demonstrates how contemporary digital systems are designed to maximize user engagement and sustain interaction through persuasive interface design. At the same time, studies of smartphone distraction suggest that these systems can fragment attention and disrupt sustained cognitive focus. In contrast, many Christian spiritual disciplines, such as prayer, scripture reading, and reflective worship, emphasize intentional attentiveness and prolonged reflection as central components of spiritual life. Although some technologies have been developed to support spiritual engagement, the broader design of digital environments often prioritizes continuous interaction rather than contemplative focus. As a result, the intersection of digital technology and Christian spiritual practice presents an important area for further exploration, particularly regarding how engagement oriented technologies may shape or influence individuals' ability to participate in spiritually reflective activities.

4 Methodology

4.1 Research Questions

This study is guided by the primary research question: How does technology affect an individual's relationship with God? To explore this question, a secondary research question is also considered: Can software application be designed to support spiritual disciplines and mitigate the distracting effects of technology on religious practice? These questions aim to investigate both the potential negative impacts of technology on spiritual life and the possibility of leveraging digital tools to enhance religious engagement. By addressing these questions, this study seeks to contribute to a deeper understanding of the complex relationship between technology and spirituality, particularly within the context of Christian practice.

To investigate these questions, this project adopts a design based approach through the development of a software prototype. Rather than solely analyzing existing technologies, this approach allows for the exploration of how specific design choices, particularly those that limit engagement and reduce distraction, may influence user behavior and patterns of interaction. The software developed in this study is intended to function as a simplified, minimalistic interface that encourages reduced screen time and more intentional device use. By examining how such a system may alter engagement patterns, this project aims to provide insight into how technological design can shape attention and, by extension, practices that

depend on sustained focus.

4.2 Hypothesis

Due to the short-term nature of the user study, these hypotheses focus on users' immediate perceptions of attention, distraction, and reflective engagement rather than long-term behavioral outcomes.

Based on the literature reviewed in HCI, smartphone distraction, and the attentional demands of Christian spiritual practices, the following hypotheses are proposed for evaluation during a controlled user testing session:

- H1: Users interacting with a minimalistic interface designed to limit engagement will report lower levels of perceived distraction compared to users interacting with a standard, engagement driven interface.
- H2: The absence of typical engagement features (e.g., notifications, personalized content) will lead to increased self-reported attentiveness and intentionality during the use of the application, as measured by post-interaction surveys.
- H3: Users will express a greater sense of reflective engagement and spiritual presence when using the minimalistic application, as compared to their typical interactions with standard digital platforms, as measured through survey responses and qualitative feedback during the user study.

4.3 Study Design

I decided to conduct a usability study using the application I built.

5 Implementation

5.1 Resources Used

The application developed for this study, God Time, was implemented using TypeScript for application logic and CSS for interface styling. TypeScript was selected to provide a structured and type-safe environment for managing application state and user interactions, improving code reliability and maintainability. CSS was utilized to create a clean, minimalistic interface that aligns with the design goals of reducing engagement and promoting intentional use.

The application was built using React Native, a cross-platform framework that allows for development across both IOS and Android devices and uses a component based architecture that promotes modularity and reusability. This choice was made for its ability to efficiently create interactive user interfaces while maintaining performance across different platforms. The development process involved iterative design and testing to ensure that the application effectively supports the intended user experience of reduced distraction and increased attentiveness during spiritual practices.

For testing and deployment during development, Expo Go was used as a supporting framework for React Native. Expo Go provides a streamlined development workflow, allowing for rapid prototyping and testing on physical devices without the need for complex native build configurations. This facilitated the iterative design process, enabling quick adjustments based on testing results.

Progress was tracked using GitHub, which provided version control and collaboration features. This allowed for organized management of code changes, issue tracking, and documentation throughout the development process. By leveraging these resources, the application was developed efficiently while adhering to the design principles aimed at fostering a more intentional and less distracting user experience.

5.2 Design Choices

A central design goal of God Time was to minimize user interaction and reduce engagement driven behaviors commonly found in modern mobile applications. Rather than encouraging prolonged use, the application was intentionally designed with limited functionality and a simplified interface to promote intentional use. This approach aligns with the study's focus on reducing distraction and supporting sustained attention during spiritual practices.

The primary feature of the application is a stillness timer, which allows users to select a set duration for engaging in spiritual disciplines such as prayer, reflection, or scripture reading. Once initiated, the timer functions without additional interaction, encouraging users to disengage from the device and focus on the intended activity. This design choice contrasts with typical engagement oriented applications that continuously prompt user interaction through notifications, personalized content, and dynamic interfaces. The timer has two modes. A standard mode that allows users to leave the app and use their phone as normal, and a strict mode that tracks if the user leaves the app and ends the timer if they do. This design choice was made to allow users to choose the level of engagement they are comfortable with while still providing an option for those who want a more restrictive experience.

In addition to the timer functionality, the application includes a reflection journal feature that a user is guided to directly after their stillness session. This feature selects a question at random from a small pool to prompt users to reflect on their experience and spiritual engagement. The journal is designed to encourage thoughtful reflection without requiring additional interaction during the stillness session itself. By separating the timer and reflection features, the application supports a focused period of spiritual practice followed by intentional reflection time to foster deeper engagement with the experience overall.

6 Results

Findings showed that...

7 Conclusion

References

[1] Finney, J. R., and Malony Jr, H. N. Empirical studies of christian prayer: A review of the literature. *Journal of Psychology and Theology 13*, 2 (1985), 104–115.

[2] Hiniker, A., and Wobbrock, J. O. Reclaiming attention: Christianity and HCI. *Interactions 29*, 4 (June 2022), 40–44.

[3] Hornbæk, K., Kristensson, P. O., and Oulasvirta, A. *Introduction to Human-Computer Interaction*. Oxford University Press, june 2025.

[4] O'Leary, T. K., Stowell, E., Sackitey, D., Yun, H. S., Wright, D., Paasche-Orlow, M., Bickmore, T., and Parker, A. G. Church after sunday: supporting everyday well-being through techno-spiritual health interventions. *Interactions 29*, 4 (June 2022), 90–93.

[5] Roche, A. *Facilitating spiritual understanding through hermeneutical and critical Bible engagement: What can be learned from the experience of a group of Christians reading the Bible with a course developed from the work of Sandra M. Schneiders?* PhD thesis, Durham University, 2020.

[6] Sims, D. B. The effect of technology on christianity: Blessing or curse. *Makalah riset, Texas Baptist University* (2001).

[7] Taylor, A., Hook, M., Carlson, J., Gudergan, S., and Falk, T. Appetite for distraction? a systematic literature review on customer smartphone distraction. *Int. J. Inf. Manag. 75*, C (Apr. 2024).

[8] Wolf, S., Moerike, F., Luthe, S., Nord, I., and Hurtienne, J. Spirituality at the Breakfast Table: Experiences of Christian online worship services. In *Extended*

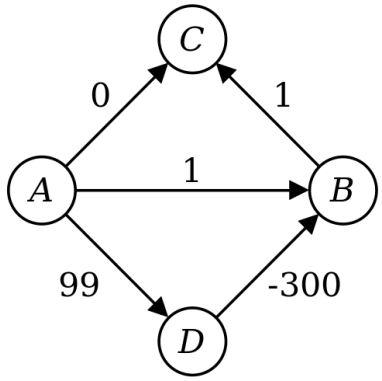

Figure 2: Dijkstra's algorithm does not work on graphs with negative edge weights, as evidenced here.

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